The Effect of Common Productivity Shock on Misallocation

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Abstract

This study investigates the impact of common productivity shocks as a macroeconomic factor influencing resource misallocation. To this end, a structural model is developed that integrates previously identified sources of misallocation -such as capital adjustment costs, partial irreversible costs of capital, variations in capital prices, financial constraints, and firm-specific heterogeneous productivity- while introducing the concept of common productivity shock, a factor that has not been examined in prior research. The results demonstrate that a doubling of common productivity at the macro level enhances output not only through direct productivity gains but also by improving allocative efficiency, reducing resource misallocation by 14%. This efficiency gain is observed specifically in economies where larger firms benefit from lower capital costs. Furthermore, over half of the observed growth is attributed to improved capital allocation, with the remainder driven by an increase in total capital stock following a 10% rise in average common productivity. The findings also indicate that as average productivity increases, the contribution of allocative efficiency to overall growth becomes more significant.

JEL Classifications: D21, D24, D25, D81, E22, G31, O47
Keywords: productivity, misallocation, TFP Loss, adjustment costs

1 Introduction

The disparity in per capita income across countries is significant, such that, according to World Bank statistics in 2022, the average per capita income of the top five countries is over 100 times that of the bottom five countries. Early studies explaining differences in per capita income primarily proposed the concept of resource growth: "Increased capital, more effective labors, and a more skilled human capital lead to economic growth." However, several studies demonstrated that differences in production inputs cannot account for a substantial portion of the per capita income gap. Thus, more recent studies introduced the concept of "resource misallocation" as a key explanatory factor: "Some countries generate higher value-added from a given level of capital and labor by efficiently allocating inputs to more productive firms." The study by Hsieh and Klenow (2009) revealed that misallocation in China and India is significantly greater than in the United States. They estimated that if input distribution in China were similar to that of the U.S., total factor productivity (TFP) in China would increase by 30% to 50%. Similarly, if input allocation in India mirrored that of the U.S., India's TFP would increase by 40% to 60%.

Following the identification of misallocation as a critical factor, a question arises: what are the sources of misallocation, through which mechanisms do they operate, and to what extent do they exacerbate misallocation? Broadly speaking, any form of heterogeneity that causes the marginal product of inputs to differ across firms generates misallocation. For example, reallocating capital from a firm with a high marginal revenue product of capital (MRPK) to one with a lower MRPK increases the overall output produced from the same amount of capital. Differences in MRPK across firms stem from various sources, including financial frictions, capital adjustment costs, differences in market power, risk and uncertainty, trade barriers, subsidies, or discriminatory taxation. Each of these sources affects misallocation through different mechanisms. This study focuses on a source that has been less addressed in the literature: "common productivity shocks" at the macroeconomic level, which affect all firms collectively. This source seems to become more prominent in economies with less stability and greater political and trade barriers.

Each of the previously identified sources affects misallocation in distinct ways: "Financial frictions" prevent more productive firms from optimally allocating resources, as they face constraints in securing the necessary capital. "Capital adjustment costs" prevent firms from instantly acquiring or selling the required capital. "Differences in market power" also contribute to misallocation. If a firm with more market power sets prices higher than its marginal cost, it will earn a higher profit margin. The prospect of higher profits attracts resources to the firm with greater market power, even if it is less productive. Thus, when firms have varying abilities to set prices, misallocation occurs. Additionally, firms take future expectations into account when deciding how to allocate

resources. Uncertainty can lead firms to make more cautious decisions. If firms are risk-averse, have less accurate forecasts, or face unexpected changes in future conditions, resources may be inefficiently allocated. Therefore, "risk and uncertainty" can be another source of misallocation. Furthermore, if the economic cost of inputs varies across firms such that more productive firms face higher costs, resources will not be optimally allocated. This can manifest in the form of trade restrictions, discriminatory taxes or subsidies, and differences in access to credit for capital acquisition.

David and Venkateswaran (2019) simultaneously considered multiple sources of misallocation and evaluated the impact of each separately. Their model, which decomposes the sources of misallocation across 11 countries, found that on average, it explains less than 50% of misallocation. Thus, a more comprehensive model was needed -one that could explain misallocation, consider the dominant mechanisms driving it, and estimate the impact of each source. In this study, common productivity shocks are introduced as a new source of misallocation. Common shocks stem from phenomena that affect all firms. These shocks may affect all firms uniformly. Estimating the impact of this source is useful for two reasons. First, it explains a significant portion of misallocation that had previously been left unexplained. Second, it refines the estimates of how each source affects misallocation.

The proposed method in this study involves using a dynamic, stochastic, partial-equilibrium structural model that incorporates various frictions in the investment and financing processes to identify firms' optimal behavior over time. By deriving the firms' policy function, this economy can be simulated to evaluate the impact of different factors on total production and misallocation.

The following chapters are structured as follows: the next chapter reviews the relevant literature on misallocation. Chapter 3 presents the structural model. Chapter 4 discusses the simulation approach, and Chapter 5 reports the results and counterfactual tests.

2 Literature Review

This research connects two main strands of the literature: studies that focus on measuring and identifying misallocation, and those that assess the effects of macroeconomic shocks . By evaluating the impact of common productivity shocks on misallocation, this study bridges the gap between these two areas of research.

The seminal study by Hsieh and Klenow (2009) played a crucial role in advancing the methods for measuring the dimensions of misallocation. They measured misallocation by considering distortions in product prices and capital prices at the firm level, without specifying a particular source[1]. Following their work, the literature moved toward iden-

[1]However, certain limitations in Hsieh and Klenow's (2009) study, such as the omission of firm-level dynamic decision-making, the lack of differentiation between industry production functions, and the

tifying specific sources of distortions and their effects on misallocation. To gain a more precise understanding, we independently examined each source and its relationship to misallocation.

Key studies in this field include Asker et al. (2014)[2], which focused on capital adjustment costs; Moll (2014)[3], which explored financial frictions; Peters (2020)[4], which examined heterogeneity in market power; and Bloom (2009) and David et al. (2016)[5], which analyzed the role of risk and uncertainty. These studies identified various factors influencing misallocation. Further contributions by Song et al. (2015), Gopinath et al. (2017), Kehrig et al. (2017), David and Venkateswaran (2019), Bento, Pedro, and Diego Restuccia (2019), and Liu et al. (2021) examined the simultaneous impact of multiple factors, accounting for the interaction of different sources in the measurement of misallocation.

Our model for evaluating misallocation incorporates several key elements: capital adjustment costs, relatively irreversible capital costs, heterogeneity in capital prices based on productivity and size, and financial frictions such as collateral constraints. These aspects are grounded in the works of Asker (2014), Cooper and Haltiwanger (2006), David and Venkateswaran (2019), and Moll (2014), respectively.

In this study, macroeconomic shocks are modeled as changes in the common component of firms' productivity. By simultaneously accounting for various sources of misallocation, we refine the analysis of their individual effects on both output and misallocation.

Our contribution lies in integrating a common productivity shock among firms as an indicator of the macroeconomic environment within a dynamic general equilibrium model that includes multiple sources of misallocation. We hypothesize that although common productivity shocks affect all firms equally, they distort the relative allocation of resources among firms. By common productivity change, we refer to long-term shifts in the productivity level of all firms, which distinguishes our work from studies such as Osotimehin (2019) that focus on the impact of business cycles on misallocation.

3 The Model

In this section, we propose a model to describe firms' decision-making processes. The model is a dynamic and stochastic partial equilibrium framework that identifies firms' optimal behavior over time, accounting for various frictions in the investment and fi-

exclusion of firm entry and exit effects, were addressed in subsequent research.

[2]In addition, Cooper and Haltiwanger (2006), Khan and Thomas (2008), Song and Wu (2015), and Kehrig and Vincent (2017)

[3]In addition, Borra et al. (2011), Gilchrist et al. (2013), Midrigan and Xu (2014), and Sunderaris and Wright (2014)

[4]In addition, Asker et al. (2019) and Baghai and Farhi (2020)

[5]In addition, Berengji and Mullainathan (2009) and Naveen Gandhi (2023)

nancing processes. Firms' productivity is divided into two components: a firm-specific component and a common component shared by all firms. The common productivity component represents the macroeconomic conditions that are uniform across firms.

In the model, households supply labor within a framework characterized by an infinite time horizon, while firms represent the demand side of the labor market. The labor supply is assumed to be inelastic, enabling firms to employ labor without restriction. Production occurs in two stages: intermediate goods are produced under monopolistic competition, and final goods are produced from the intermediate goods in a competitive market. The firm's problem is then described in detail.

Firms face four types of frictions in the investment process. First, the cost of maintaining capital varies across firms and depends on their level of capital, productivity, and firm-specific characteristics. Additionally, firms encounter capital adjustment costs, which are convex functions of investment, as well as relatively irreversible costs associated with changes in capital, which are nonlinear and represent a form of price rigidity. The fourth friction is a constraint on the maximum amount of capital a firm can raise, limiting firms to invest only up to a fixed proportion of their output. This represents a quantity constraint.

3.1 Production Functions and Market Structure

Production in this economy occurs in two stages. In the first stage, a continuum of firms, indexed by i, produces intermediate goods according to the production function described by Equation 1:

$$Y_{it} = K_{it}^{\hat{\alpha_1}} N_{it}^{\hat{\alpha_2}}, \qquad \hat{\alpha_1} + \hat{\alpha_2} = 1 \tag{1}$$

where K_{it} is the capital stock, and N_{it} is the labor employed by the firm at time t. In the second stage, a final goods producer uses a Constant Elasticity of Substitution (CES) aggregator to combine intermediate goods and produce the final good, as represented in Equation 2:

$$Y_t = \left(\int \hat{A}_{it} Y_{it}^{\frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}} \tag{2}$$

where θ is the elasticity of substitution between intermediate goods, and $\hat{A}_{it}$ is the firm-specific productivity level. The final good is produced in a frictionless competitive market. The demand for the intermediate goods producer can then be derived from this setup as follows in Equation 3:

$$\max P_t Y_t - \int P_{it} Y_{it} di = \max P_t \left(\int \hat{A}_{it} Y_{it}^{\frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}} - \int P_{it} Y_{it} di \tag{3}$$

$$\frac{P_{it}}{P_t} = \left(\frac{Y_{it}}{Y_t}\right)^{-\frac{1}{\theta}} \hat{A}_{it} \tag{4}$$

where P_{it} is the price of the intermediate good produced by firm i, and P_t is the aggregate price level of the final good. This equation captures the demand for each intermediate good as a function of its relative price and firm-specific productivity.The price of the final good, P_t, is normalized to 1, as it serves as the numeraire good. Given this normalization, the demand function for the intermediate firm becomes as shown in Equation 5:

$$P_{it} = \left(\frac{Y_{it}}{Y_t}\right)^{-\frac{1}{\theta}} \hat{A}_{it} \tag{5}$$

Using this demand function, the revenue of each intermediate firm can be calculated as follows in Equation 6:

$$P_{it}Y_{it} = \left(\frac{Y_{it}}{Y_t}\right)^{-\frac{1}{\theta}} \hat{A}_{it}Y_{it} = Y_t^{\frac{1}{\theta}} \hat{A}_{it} {Y_{it}}^{1-\frac{1}{\theta}} \tag{6}$$

Next, by substituting the firm's production function from Equation 1, the firm's revenue can be expressed as shown in Equations 7 and 8:

$$P_{it}Y_{it} = Y_t^{\frac{1}{\theta}} \hat{A}_{it} \left(K_{it}^{\hat{\alpha_1}} N_{it}^{\hat{\alpha_2}}\right)^{1-\frac{1}{\theta}} = Y_t^{\frac{1}{\theta}} \hat{A}_{it} K_{it}^{\alpha_1} N_{it}^{\alpha_2} \tag{7}$$

$$\alpha_j = \left(1 - \frac{1}{\theta}\right) \hat{\alpha_j}, \qquad j \in \{1, 2\} \tag{8}$$

3.2 Labor Costs

In this model, households supply labor in an infinite-horizon economy, while firms demand labor. Since labor is supplied inelastically to the firms, they can hire as much labor as they deem necessary without facing any frictions. The wage rate for labor, W_t, is constant across all firms.

3.3 Capital and Investment Costs

3.3.1 Capital Adjustment Costs

Investment is subject to quadratic capital adjustment costs, which are modeled as follows in Equation 9:

$$\Phi_A(K_{it+1}, K_{it}) = \frac{\xi}{2} \left(\frac{K_{it+1}}{K_{it}} - (1 - \delta)\right)^2 K_{it} \tag{9}$$

3.3.2 Partially Irreversible Costs of Capital

The gap between the purchase and sale price of capital introduces costs for firms when they buy and later sell capital. The partial irreversibility of capital stems from transaction costs, the lemons problem, and the physical costs associated with resale. When new capital is installed in a firm, there may also be temporary productivity losses, such as factory downtime for refurbishment or slower production processes. These costs are modeled as a fraction of the investment. Equation 10 represents the irreversibility cost function:

$$\Phi_N = \phi_n(P_{gap}\mathbb{1}(I > 0) - \mathbb{1}(I < 0)) \mid I \mid \tag{10}$$

In other words, we have the following for different cases:

$$\Phi_N = \begin{cases} \phi_n P_{\text{gap}} I & \text{if } I > 0 \\ 0 & \text{if } I = 0 \\ \phi_n \mid I \mid & \text{if } I < 0 \end{cases} \tag{11}$$

where ϕ_n is the parameter representing the irreversibility cost, P_{gap} is the price gap between buying and selling capital, and I is the firm's investment level. This formulation captures how positive investment incurs costs, while negative investment -representing disinvestment- also faces penalties, though of a different nature.

3.3.3 Cost of Capital

In addition to capital adjustment costs, other factors influence the final cost of capital for firms. These factors may include government interventions and regulations, such as subsidies or discriminatory taxes on firm capital or financial constraints. All of these are modeled as a tax on the firm's capital cost flows, represented by the term T_{it}^K, which reflects the total cost of capital acquired by the firm. If firm i decides to increase its capital at time t, it must pay T_{it}^K per unit of capital. In other words, the price per unit of capital equals T_{it}^K. The cost of investment is modeled as follows to capture the effect of various factors on capital pricing:

$$\log T_{it}^K \equiv \tau_{it} = \gamma_a a_{it} + \gamma_k k_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2) \tag{12}$$

In the equation above, the parameter γ_a represents the correlation between capital price shocks and firm productivity (which includes both firm-specific and common components of productivity), while γ_k captures the correlation between capital price shocks and the firm's capital stock (firm size). The term ε_{it} is a purely random component. This formulation allows us to account for how productivity and firm size influence the cost of acquiring capital, in addition to the random shocks that affect capital prices.

3.4 Financial Constraints

If a firm faces financial constraints in securing capital, for example, being unable to obtain the necessary capital through loans, it may be unable to reach its optimal capital stock. This constraint is incorporated into the model as the inequality presented in Equation 13:

$$K_{it+1} \leq K_{it}(1-\delta) + \xi A_{it} K_{it}^{\alpha} \tag{13}$$

where ξ is the financial accessibility parameter, indicating how much the firm can invest as a multiple of its output.

3.5 Productivity Dynamics

This section describes the stochastic process governing firm productivity shocks (a_{it+1}). In this study, we consider both idiosyncratic and common productivity shocks across firms. The productivity shock consists of a common component, m_t, and a firm-specific component, z_{it}. The common component is constant across firms but varies over time. Each component follows a first-order autoregressive process. The stochastic processes for productivity are shown below:

$$a_{it} = z_{it} + m_t \tag{14}$$

$$z_{it+1} = \rho_z z_{it} + (1-\rho_z)\bar{z}_i + \mu^z_{it+1}, \quad \mu^z_{it} \sim N(0, \sigma^2_{\mu^z}) \tag{15}$$

$$m_{t+1} = \rho_m m_t + (1-\rho_m)\bar{m} + \mu^m_{t+1}, \quad \mu^m_t \sim N(0, \sigma^2_{\mu^m}) \tag{16}$$

3.6 Firm's Problem

In each period, labor is chosen instantaneously and without friction, meaning the firm can hire the optimal amount of labor by paying the competitive wage W_t. The firm's problem is represented by the Bellman equation in the following form:

$$\begin{aligned} V(K_{it}, N_{it}, A_{it}) = \max \mathbb{E}_{it}\Big[& Y_t^{\frac{1}{\theta}} \hat{A}_{it} K_{it}^{\alpha_1} N_{it}^{\alpha_2} - W_t N_{it} - T^K_{it+1} K_{it+1} \\ & + T^K_{it} K_{it}(1-\delta) - \Phi_A(K_{it+1}, K_{it}) - \Phi_N(K_{it+1}, K_{it}) \\ & + \beta V(K_{it+1}, N_{it+1}, A_{it+1})\Big] \end{aligned} \tag{17}$$

where E_{it} represents the firm's expectations at time t, and the firm optimizes the amount of labor, N_{it}, in each period. After optimizing over labor, substituting the optimal

labor choice into the firm's problem, and redefining the value function, the Bellman equation simplifies to:

$$\begin{aligned}\mathcal{V}(K_{it}, A_{it}) = \max E_{it}\Big[& GA_{it}K_{it}^{\alpha} - T_{it+1}^{K}K_{it+1}(1-\beta(1-\delta)) \\ & - \Phi_A(K_{it+1}, K_{it}) - \Phi_N(K_{it+1}, K_{it}) \\ & + \beta\mathcal{V}(K_{it+1}, A_{it+1})\Big]\end{aligned} \tag{18}$$

$$G \equiv Y_t^{\frac{1}{\theta(1-\alpha_2)}} \left(\frac{\alpha_2}{W_t}\right)^{\frac{\alpha_2}{1-\alpha_2}} (1-\alpha_2) \tag{19}$$

$$A_{it} \equiv \hat{A}_{it}^{\frac{1}{1-\alpha_2}} \tag{20}$$

$$\alpha \equiv \frac{\alpha_1}{1-\alpha_2} \tag{21}$$

In this equation, G represents the influence of macroeconomic variables, and α captures the curvature of the profit function. A_{it} denotes the profitability of physical capital for each firm, which is referred to simply as productivity in subsequent sections. For simplicity, G has been normalized to 1. It is important to note that if the production cost coefficients are identified, from a modeling perspective, T_{it+1}^{K} reflects the relative productivity of capital costs compared to output.

3.7 Misallocation

To measure the extent of misallocation, the variance of the marginal revenue product of capital (MRPK), denoted as σ^2_{mrpk}, is used. MRPK is the logarithm of the derivative of the production function with respect to capital, which is obtained from the following relationship:

$$\text{mrpk} \equiv \log\left(\frac{\partial f(K_{it}, A_{it})}{\partial K_{it}}\right) = \alpha A_{it} K_{it}^{1-\alpha} \tag{22}$$

The variance of this indicator across firms represents the differences in marginal productivity. If mrpk differs across firms, reallocating one unit of capital from a less productive firm to a more productive one would increase total output. Therefore, σ^2_{mrpk} serves as an index of the degree of capital misallocation among firms in the economy.

4 Simulation

4.1 Policy Function Calculation

The firm's policy function is computed numerically using the Value Function Iteration (VFI) method. This approach involves starting with an initial guess for the value function, which is then used in the optimization of the Bellman equation to generate a new value function. This updated function is iteratively used in subsequent optimizations, and by the contraction mapping theorem, the process converges to the optimal value function after several iterations. Once the value function converges, the policy function can be derived, which dictates the optimal decisions for capital and productivity levels at each point in time.

In figure 1, the policy function is plotted for five different productivity levels. The locus of constant capital is also plotted as a dashed line, represented by $k' = k(1 - \delta)$. If the policy function lies above (below) this line, the firm will increase (decrease) its capital stock. Due to adjustment costs associated with capital changes, firms may prefer to make no changes to their capital stock in the neighborhood of this constant-capital line, investing only enough to offset depreciation. This region where the policy function adheres closely to the constant-capital line is referred to as the inaction set. In this "inaction region," firms avoid adjusting their capital due to the costs of capital adjustment, preferring to maintain their current capital stock unless there is a significant incentive to deviate from this strategy.

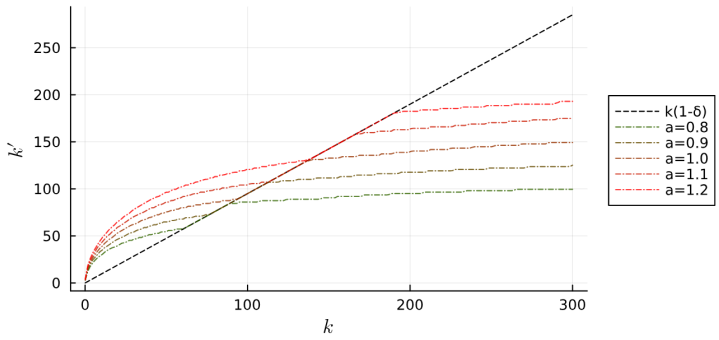


Figure 1: Firm's Policy Function

In figure 2, the contours of iso-investment can be observed. As seen from the graph, the higher a firm's productivity, the more it will invest. Consequently, the inaction set, where the firm chooses not to adjust its capital stock, shifts to a higher level of capital.

4.2 stationary distribution

Using the policy function, the firm's future decisions can be determined in every state. Thus, it is possible to track the firm's decisions over time. By simulating the firm's

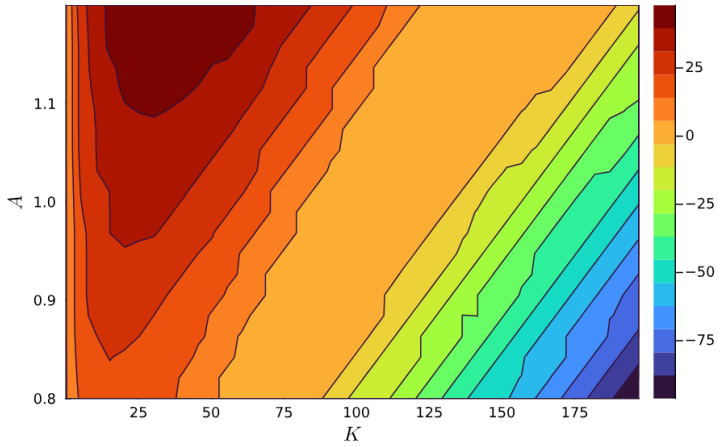


Figure 2: Iso-Investment Curves

decisions over $M = 10,000 + 100$ periods and discarding the first 100 periods, we can compute the stationary distribution of variables such as capital, investment, firm-specific productivity, common productivity, capital price, and MRPK.

This process allows for a more precise identification of the steady-state levels and the distributions of these variables, which are crucial for understanding the long-term equilibrium behavior of the firm in the absence of shocks or capital adjustment frictions.

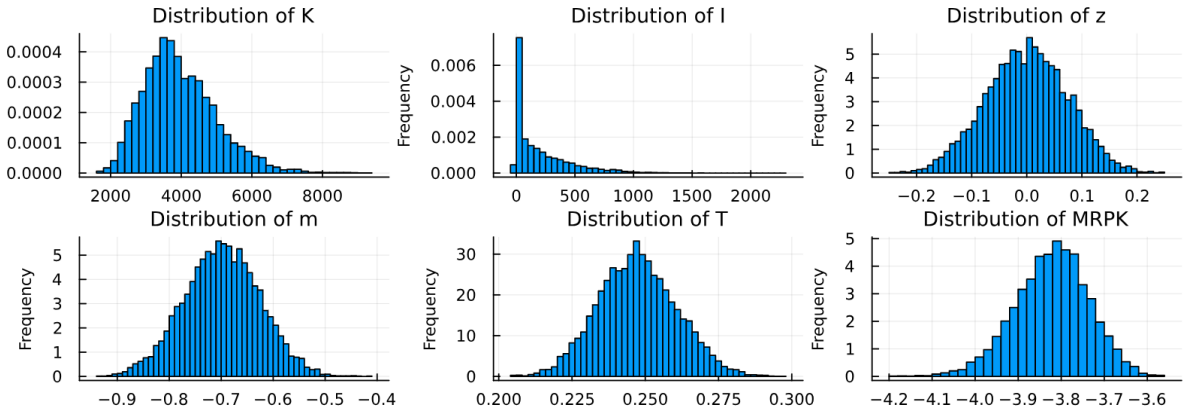


Figure 3: The steady-state distribution of capital (K), investment (I), firm-specific productivity (z), common productivity (m), capital price (T), and MRPK for a representative firm

4.3 Aggregate Indicators

After simulating firms over time, total production, total capital, and total investment are obtained by aggregating the corresponding firm-level variables at each time period. Additionally, misallocation is quantified by calculating σ^2_{mrpk} at each time step. To provide a more precise statistical analysis, the mean and standard deviation of these indicators are computed over time and reported in the table below.

Table 1: Aggregate Indicators in Baseline Model

Title	K_total	I_total	Y_total	MRPK_var
Value	90,944,010	4,546,038	2,025,580	0.05562
Std	14,968,181	3,389,044	341,814	0.00167

This table summarizes the dynamic behavior of the economy over the simulation period, capturing the trends in production, capital accumulation, investment patterns, and the extent of misallocation among firms.

5 Simulation

Upon conducting a comprehensive simulation of the model, it becomes feasible to evaluate the influence of various parameters on firm behavior. Among the most critical determinants of resource misallocation is the average common productivity $\bar{m}$. Misallocation metrics are computed across a spectrum of average common productivity values, as depicted in Figure 4 . The baseline model suggests that in economies characterized by elevated levels of common productivity, misallocation tends to be reduced. This relationship implies that enhancements in macroeconomic productivity not only augment allocative efficiency but also contribute to production growth, driven by both heightened productivity levels and optimized resource allocation.

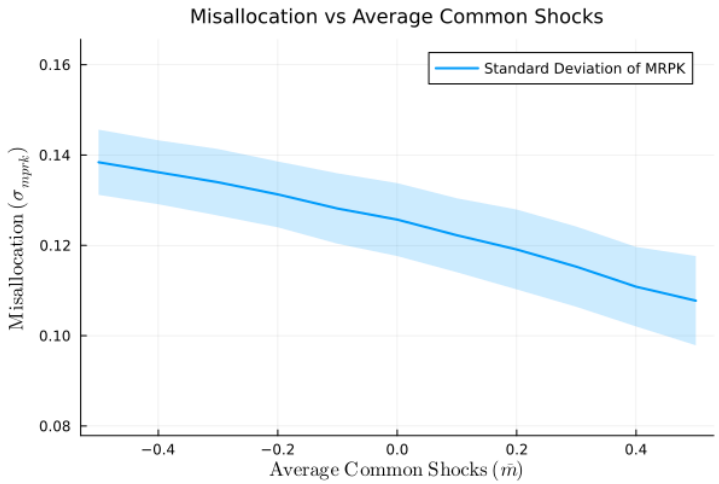


Figure 4: Misallocation vs Average Common Productivity

5.1 Disentangling the Effects of Capital Growth from Capital Allocation

An increase in average productivity across firms induces dual alterations in the capital structure of the economy. First, the augmentation of total productivity leads to a rise in

the aggregate capital stock. Second, there is a shift in the allocation of capital among firms. Consequently, the growth resulting from heightened common productivity can be attributed to two distinct sources. The following methodology delineates a framework to disentangle these sources by examining an intermediate state.

Let us denote the economy's initial state as i, wherein each firm possesses a specific quantum of capital. By aggregating the capital held by each firm, we derive the vector K_i, which encapsulates the total capital. The corresponding output generated by these firms can be expressed as the vector Y_i. Upon the occurrence of a positive shock, the economy transitions from state i to a final state f, wherein the capital configuration is represented by the vector K_f. This transition signifies an increase in both the total capital stock and the proportional distribution of capital among firms.

To effectively disentangle these two phenomena, we establish an intermediate state in which the total capital is equivalent to that observed in the final state, yet the distribution mirrors that of the initial state. Mathematically, this is articulated as vector K' aligning with vector K_i while maintaining parity in magnitude with vector K_f. Utilizing the capital configuration in this intermediate state alongside the production function of each firm, we can compute the output of each firm, denoted as vector Y'.

By contrasting the total output in the intermediate state with that in the initial state, we can isolate the growth attributable solely to capital expansion. Moreover, a comparison between the final state and the intermediate state facilitates the assessment of growth derived from the reallocation of capital. Figures 5 and 6 elucidate this framework, illustrating the process of disentangling growth attributable to capital size and allocation, as well as depicting the contribution of each factor in relation to average common productivity.

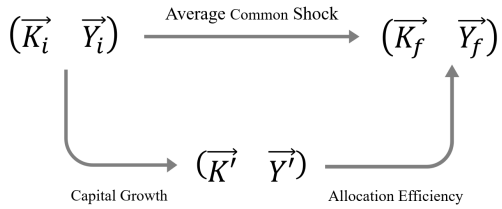


Figure 5: How to separate the growth caused by the size and allocation of capital

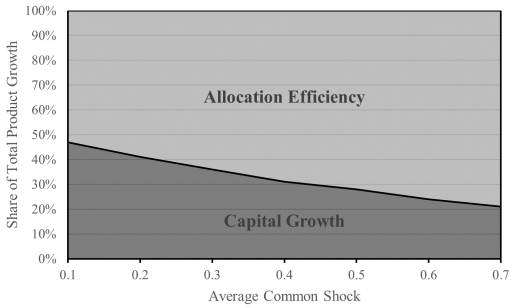


Figure 6: the share of each source in the growth in terms of average common productivity

The findings reveal a compelling relationship between the average shock to general productivity and the significance of capital reallocation in driving economic growth. Specifically, as the magnitude of this shock increases, the role of capital reallocation in facilitating growth becomes increasingly critical. When total capital expands propor-

tionately without a corresponding adjustment in the allocation of capital shares among firms relative to their productivity, a substantial portion of potential growth remains unrealized. Furthermore, the greater the enhancement in common productivity, the more pronounced the adverse effects of misallocation on overall productivity become.

This underscores the vital importance of efficient capital allocation in effectively leveraging the benefits derived from productivity improvements. Absent measures to rectify misallocation, the prospective gains from heightened productivity are unlikely to be fully harnessed. Consequently, this highlights the urgent need for policies aimed at fostering optimal resource distribution among firms to maximize the impact of productivity advancements.

5.2 Counterfactuals

To evaluate the impact of various types of frictions on misallocation, a model was executed in the absence of any frictions, with the findings illustrated in the figure below. The results indicate that if the average productivity is halved, misallocation will increase by 14%.

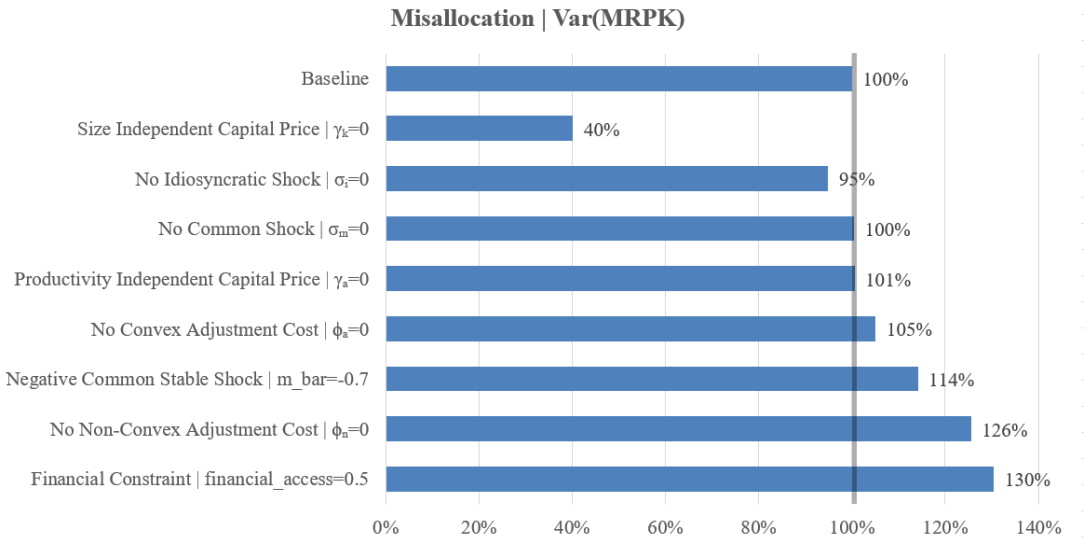


Figure 7: Misallocation Counterfactuals

5.3 Future Research Directions

Future research is encouraged to calibrate the model parameters using advanced techniques, such as the Generalized Method of Moments (GMM). Rather than estimating all parameters concurrently, a more nuanced approach could involve the identification of certain parameters independently. Furthermore, rather than adopting a generalized approach to common shocks, future work could delve deeper into the specific mechanisms through which distinct common shocks -such as monetary or fiscal policy- affect the economy. A more granular exploration of the effects of heterogeneity in market power and

production technologies should also be considered. In addition, the influence of labor market frictions and informational frictions on firms, particularly in response to both specific and common productivity shocks, warrants thorough investigation. There is also significant potential in exploring the role of managerial foresight within models, specifically by considering finite-horizon scenarios as opposed to infinite-horizon frameworks. This shift could also accommodate the introduction of Stochastic discount factor, enriching the analysis of decision-making under uncertainty and the long-term impact of productivity shocks.

6 Conclusion

This study examines the impact of common productivity shock as a macroeconomic phenomenon on resource misallocation. To this end, a structural framework is presented that, in addition to considering previously identified sources of misallocation -such as capital adjustment costs, partially irreversible costs of capital, differences in capital prices, financial constraints, and firm-specific productivity differences- also incorporates "common productivity shock", a factor that has not been previously addressed.

The findings reveal that a doubling of common productivity at the macro level not only increases output through the productivity channel but also improves allocative efficiency, reducing resource misallocation by 14%. This improvement in allocative efficiency occurs exclusively in economies where larger firms face lower capital costs. Furthermore, it was shown that when the average common productivity shock reaches 0.1, more than half of the growth is attributed to improved capital allocation, while the remainder is due to an increase in the overall capital stock. Additionally, as average productivity increases, the contribution of allocative efficiency to growth also rises.

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